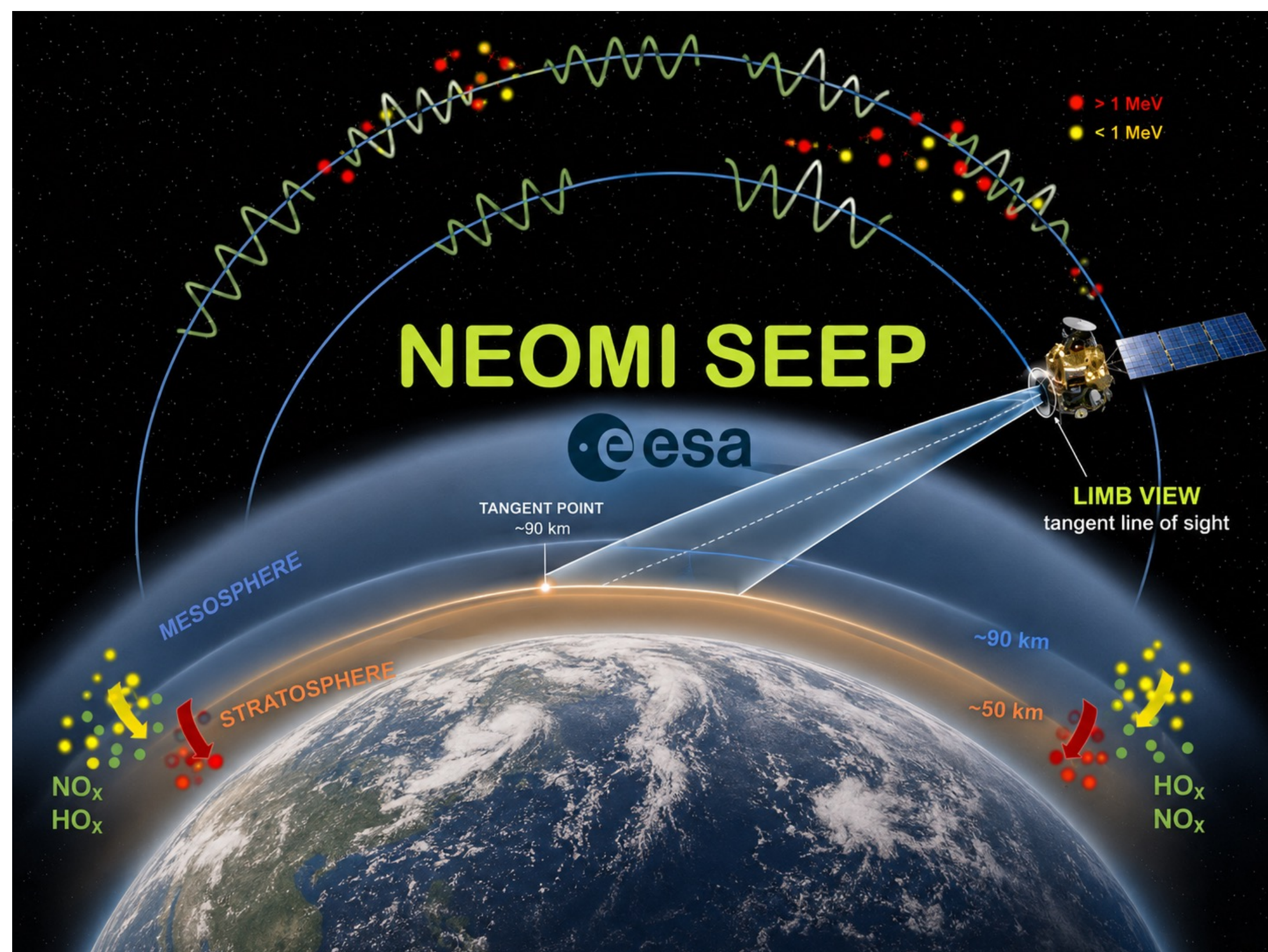


Study of Energetic Particle Precipitation and Atmospheric Impacts (SEEP): An ESA New Earth Observation Mission Idea (NEOMI)

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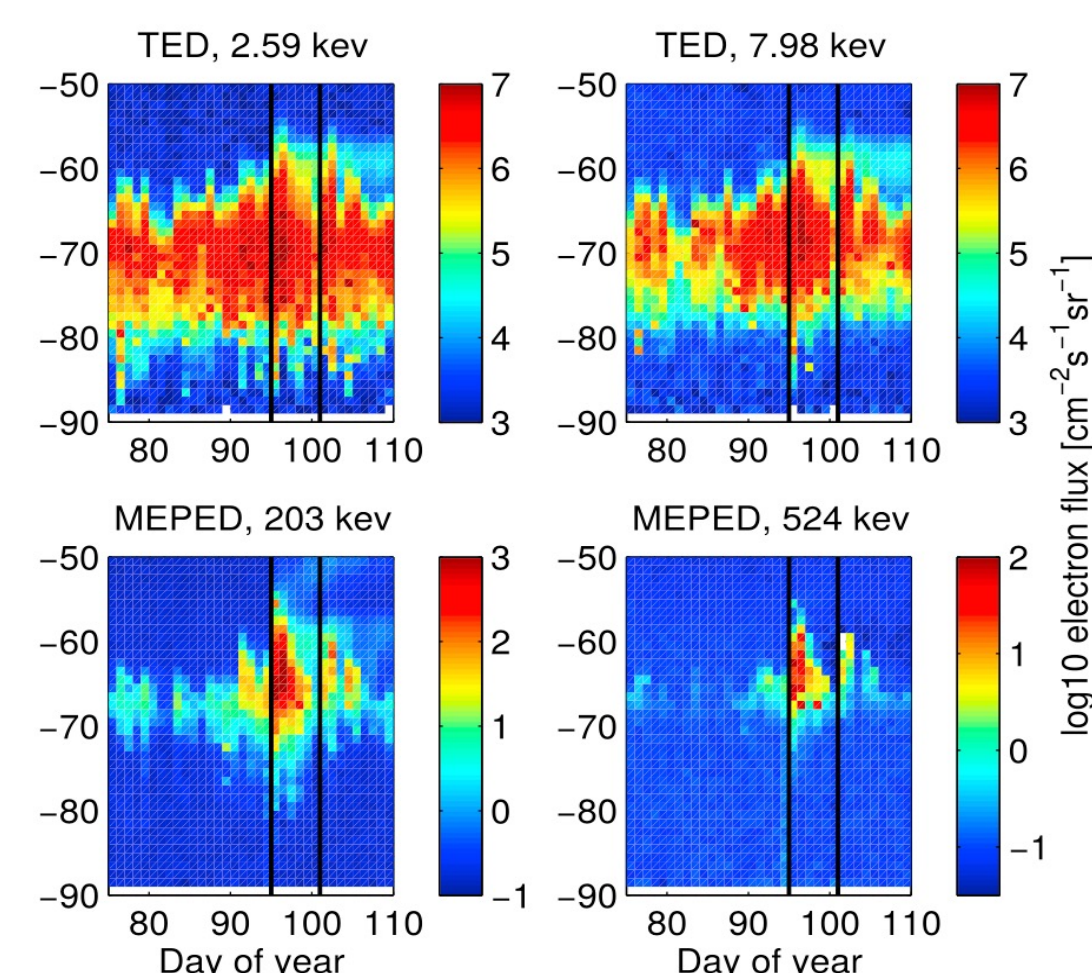
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1. Key Science Questions



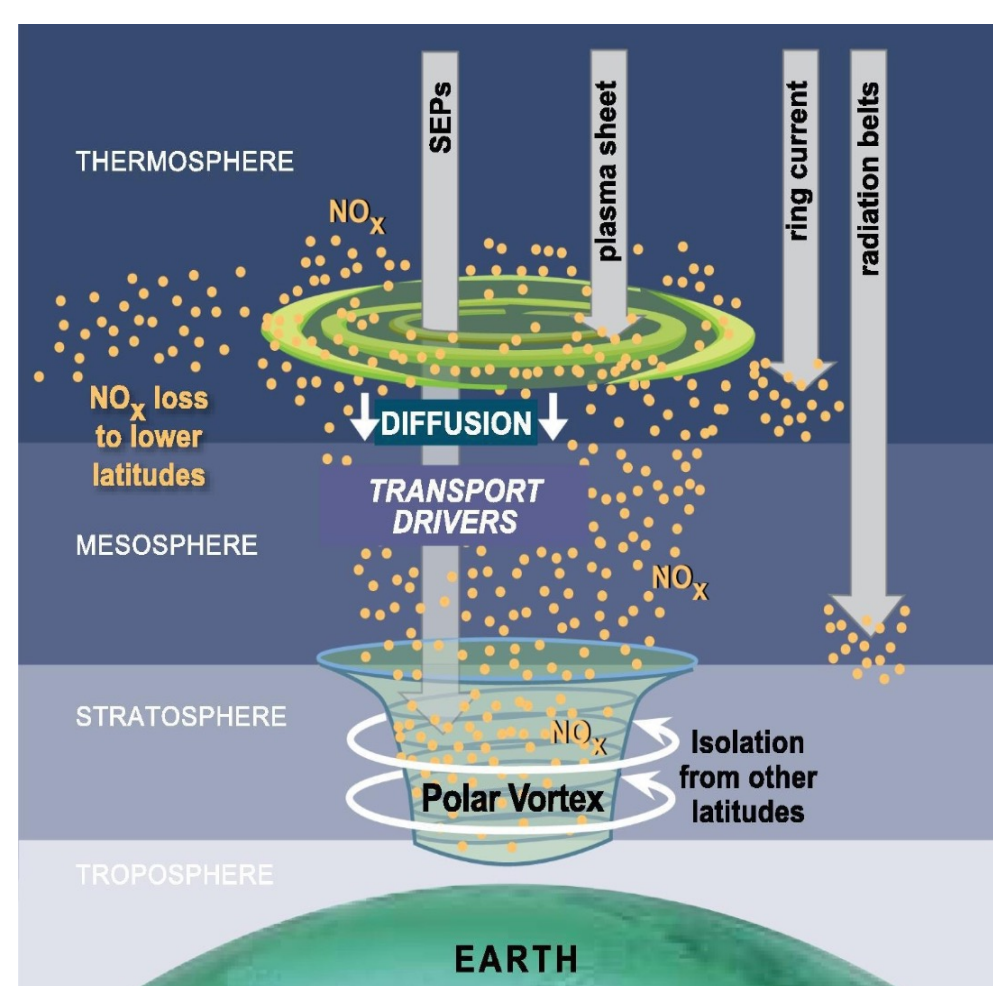
SQ1 — EPP forcing and atmospheric impact:
How much forcing enters the atmosphere through energetic particle precipitation (EPP), and what direct compositional changes does it produce?

Discrete energy channels (TED/MEPED) lead to non-continuous coverage, limiting accurate reconstruction of total energy input, from Smith-Johnsen et al. (2017).

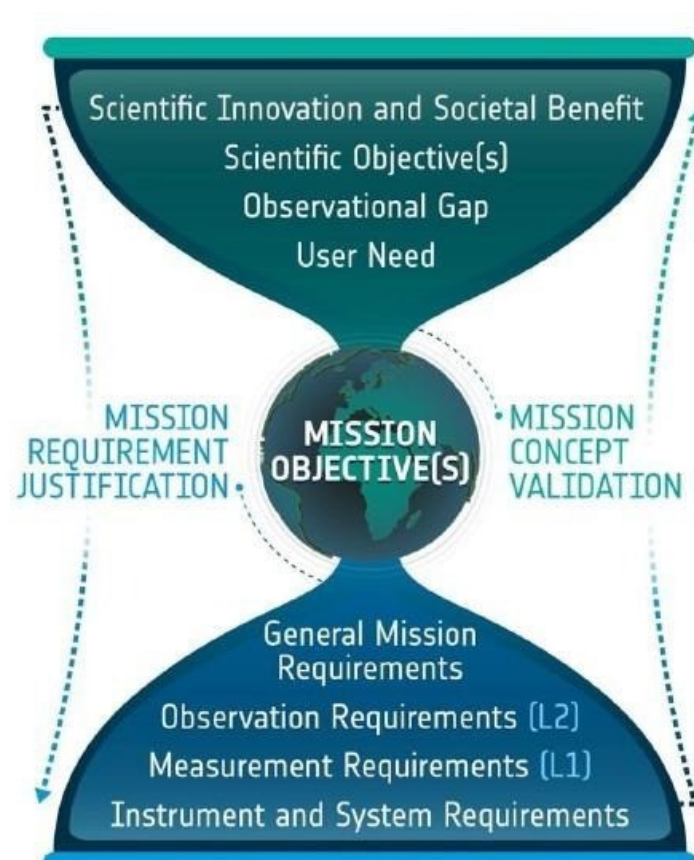


SQ2 — Transport, coupling, and climate variability:
Beyond the direct chemical effects, how are atmospheric dynamics and coupling processes altered, and what indirect responses do they generate?

[Marshall et al, 2020, Advances in Space Research]

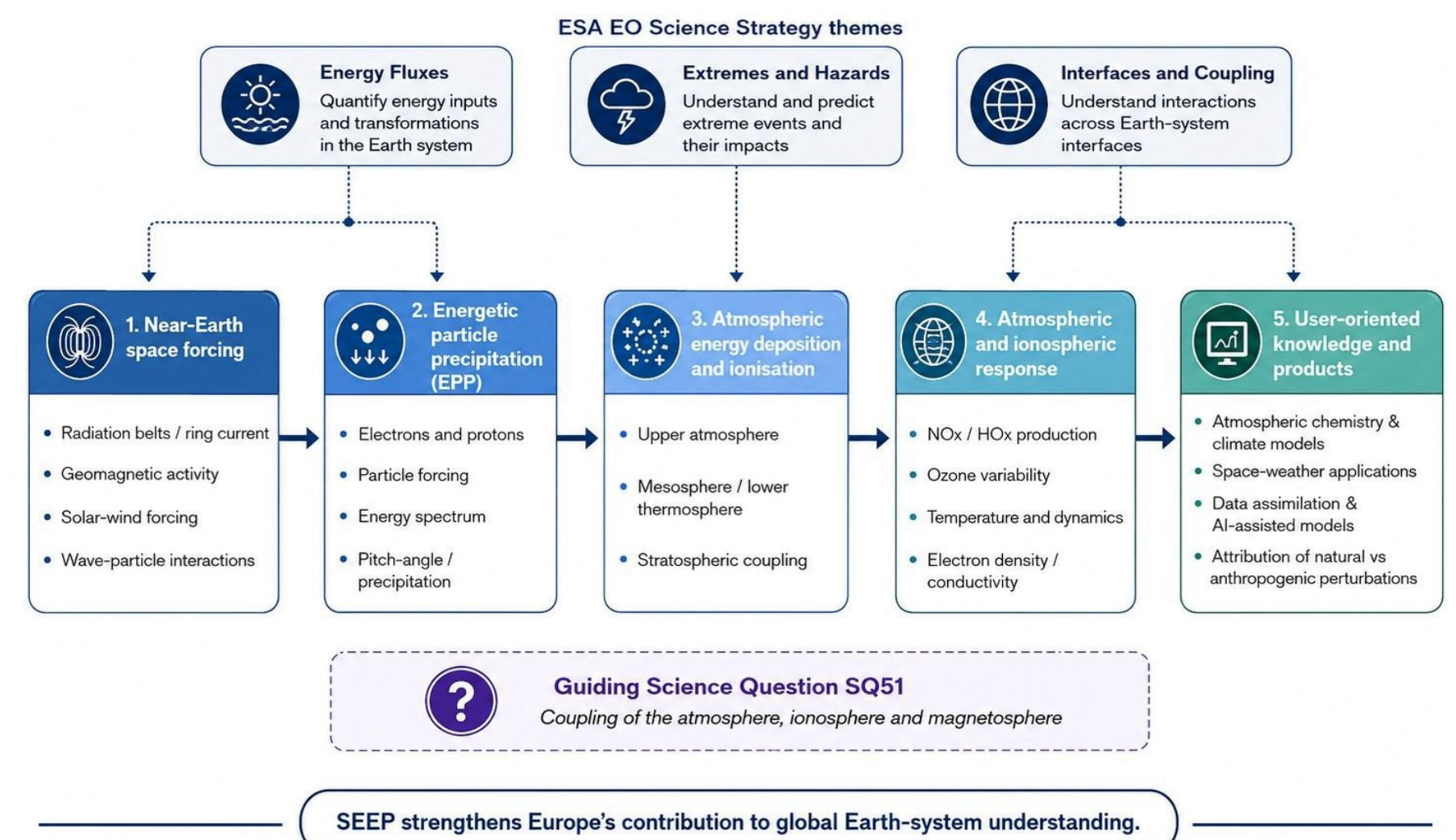


SQ3 — Modelling, prediction, and long-term impacts:
What are the impacts of EPP forcing on atmospheric and climate-relevant processes, and how can they be better represented in models?

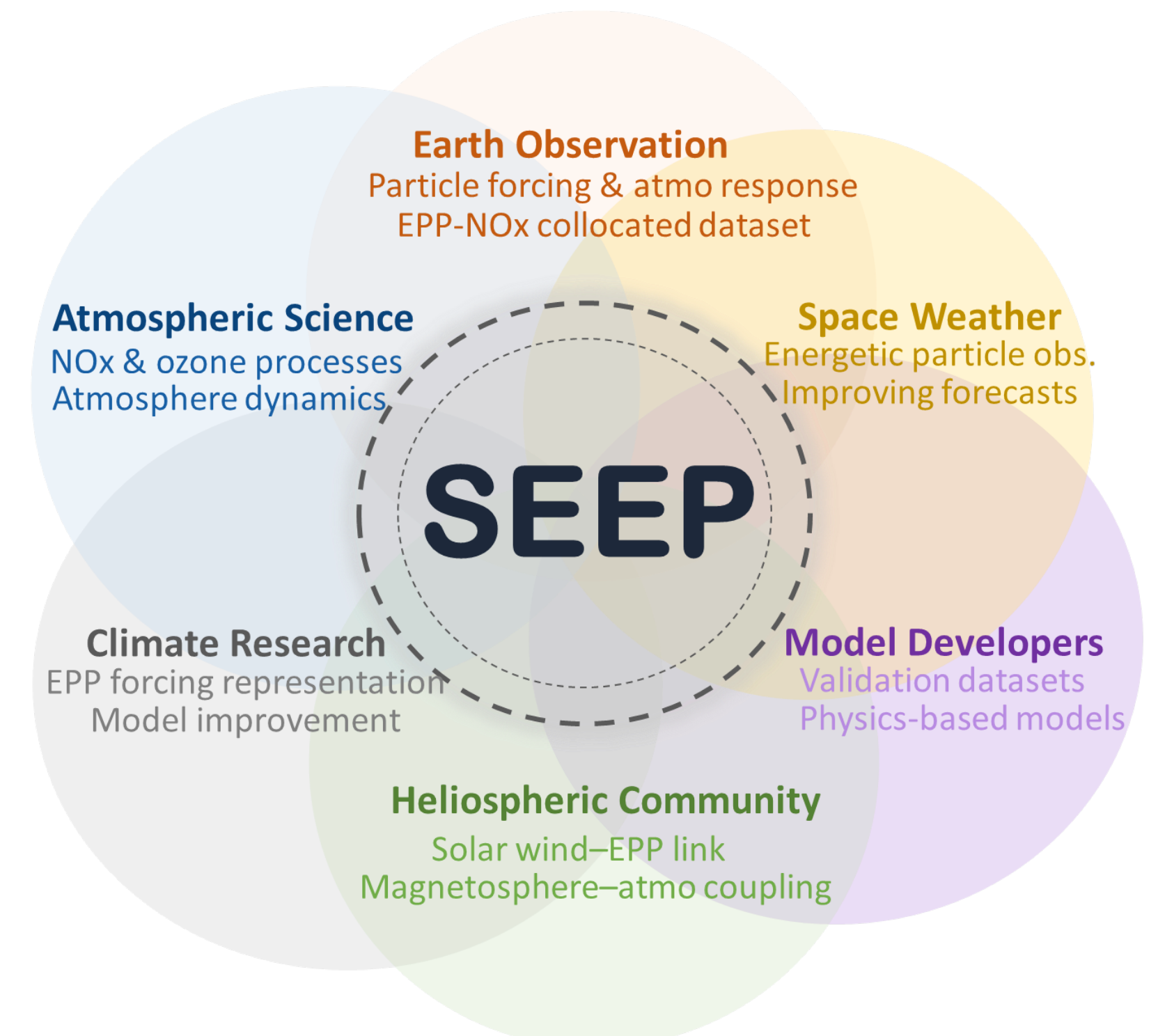


Mission objectives and requirements. (source: ESA, NEOMI Statement of work)

2. Traceability to the ESA EO Science Strategy



3. Who Benefits from SEEP Observations?



Summary and Discussions

- ✓ Current satellite measurements sample particle precipitation at limited energies and pitch-angles, which limits our ability to estimate the total energy input into the atmosphere.
- ✓ SEEP will overcome this by providing continuous, gap-free measurements of electrons and protons across a wide energy range, combined with physics-based modelling to derive atmospheric energy deposition.
- ✓ SEEP combines particle, chemical, and temperature observations to quantify energetic particle forcing on the atmosphere.

Project Website



Share your input



We invite you to provide feedback on observational needs, data gaps, and priority applications relevant to the NEOMI SEEP mission idea.